

Tri-Star Falcon: UAV with VTOL Stability and Speed Flight Modes.

Abstract- In this paper we will discuss about a newly designed model of an UAV which has been specially made for Vertical Take-off & Vertical Landing (VTOL) and as well as a mode which converts it to a high speed flight mode UAV which has been never been designed yet with a better feasibility factor. We have worked on the mechanical design now which stands to be feasible as it being tested on the aerodynamic bed as the mechanical design has been tested well. More from the mechanical side, describing the features of the UAV, it had fail safe modes like, if the drone is with one damaged motor, then the back or front part actuators can individually bring the drone back to base station acting as the backup and main drive. And apart from mechanical side, it has features like GPS tracking, live video streaming, High point stability, Stealth modes and more. These all come to the next phase of the project. In brief, in this abstract we'll discuss about some of the mechanical design and its description.

Keywords- UAV, VTOL, TSF, Control loops, autopilot, surveillance, System kinematics and dynamics.

I. INTRODUCTION

The unmanned aerial vehicle (UAV) is a small plane which flies in an autonomous way with only one virtual pilot on board. That does not mean that the plane is completely out of control once it leaves the track. The presence of a human being in the loop of control is essential. But the operator is on a station on the ground, either on ground, or at sea or in a control center. He can intervene at any time, examine the situation and the rules of engagement, and stop the mission if necessary. The concept of an UAV resonates in several research laboratories of any developed country. The first instigators of this type of the projects were the soldiers. Indeed, the armies of many countries prefer to rather send UAV's for the recognition in the high-risk zones than man. This type of flying machine can take much various missions, extending from the control of the scientific experiments to the intelligence collecting the monitoring during the day or the night. The numerical simulation in the field of aerodynamics is relatively recent research tools. The specialists raise the question of knowing if the field of aerodynamics is purely theoretical or numerical, they estimate however that it is theoretical, because of the many numerical tests necessary for the stage to insufficiencies of the knowledge of the calculation methods. Since a score of years, one can say that numerical aerodynamics lost much of its empirical nature due to the convergent efforts of mathematics to solve the nonlinear problems involved in the calculation methods in aerodynamics and also thanks to the invention of the very fast machines of calculation and the development of data processing.

In this present work, a graphical simulation of a force around an UAV and description of the flight dynamics and its control system is presented. The first phase of this work consists in using plans of construction of this UAV to

create a solid model which specifies its external geometry in Solid Works. This solid model is then exported in to a program like Autodesk Inventor of the grid generation in order to return its geometry adapted to aerodynamic calculations and tested in various flying conditions. A computer code of computational fluid dynamics (CFD) is used to obtain estimates of the coefficients of lift and drag to 10 meters per second. A study of the independence of the grid is then carried out to determine the precision of the grid of calculation used for analysis CFD and other airfoil stress simulations. Secondly the flight dynamics is put into account where necessary data is extracted for the control system to act according to the attitude of the flight.

II. ADVANCEMENTS

In normal UAVs like the Reaper and Panther it contains the universal flight mode but lacks fail safe modes of landing at high terrain places during emergency thus weakening their part. Now combining both the features brings to an ultimate UAV which proves a lot during the war field. But it's not that simple to design such a model as it need a lot of mechanical factors which falls good during aerodynamic conditions as a small force point at the non-pressure points will lead to damage and crash landing. There have been jet fighters like the Jet Harrier with VTOL and Flight modes but they are for attack. And there have been a lot of designs for Surveillance with both these features but no one has yet brought down to a feasible and marketable application. We have worked on the mechanical design now which stands to be feasible as it being tested on the aerodynamic use as the mechanical design has been tested well. The aerodynamic and mechanical simulation are described here which will prove to its feasibility. More from the mechanical side, it can stop very swiftly during flight modes in the case of emergency. It does not have to take a gliding turn when needed, it just needs to stop at the point and turn at the point itself thus saving the flight time. In brief, in this abstract we'll discuss about some of the mechanical design, and its simulation and analysis. The photos are provided for a well virtual creation and analysis.

III. CONCEPT OF OPERATION

This Concept of Operations (CONOPS) describes the operational employment of the Unmanned Aerial Vehicles (UAVs) being designed for short range, low altitude and low endurance surveillance. This CONOPS will provide an overview of the principal UAV components and organization, the intended operational environment, and the primary command and control relationships and responsibilities. It also provides a framework for the development of theater-specific concepts of employment and operations planning documents.

A. UAV Features

The UAV is designed for short range, low altitude and short endurance surveillance missions in urban and non-urban environment. The primary objective of the UAV is to automatically navigate to the target based on the waypoint commands, provide video images of the target in real time and to return back to base without any assistance. The image is captured through an 8MP video camera and is transmitted in real-time to the operator at the Ground Control Station. The operator can identify features like target movement, target location and nature of the target. These UAVs will be deployed to support forces in worldwide peace, crisis, and wartime operations such as: military surveillance and reconnaissance; urban crowd control; hostage situations etc. The high mobility of these vehicles and their accompanying ground-based system components will allow flexibility in selection of operating locations and employment. This capability will enhance the systems utility, survivability and readiness. The UAV can also be integrated with two more cameras for obstacle avoidance capability. The aircraft will have a maximum range of 50km and a top speed of 20 m/s, assuming a target surveillance duration of 10 minutes, the UAV should have a maximum endurance requirement of 90 minutes including a maximum time to travel to the target and back. The total system is compact, easy to setup and mobile with a weight of less than 10 kg. The UAV has a 11.2volt, 6000 mAh lithium polymer battery for its motor and actuators and a 7.4 volt, 1000mAh battery for its autopilot system. The complete UAV system consists of the following

1. The Unmanned Air vehicle: The UAV is an integrated system consisting of the airframe, the avionics and the payload.
2. The Ground Control Station: The Ground control station is the system which controls and monitors aircraft operations. It receives real time aircraft data and video signals from the UAV. The operator can preset waypoint and/or set waypoints during flight. The hardware used is a rugged portable computer with preloaded software.
3. Launch and Recovery System: The Launch system is simple and easy to setup device used to accelerate and launch the UAV consistently.

B. The Operation

The UAV comes as a ready to deploy, plug and play system. The autopilot software is pre-installed in the onboard hardware and the waypoints and mission specifications can be loaded into it from the ground control station (GCS) using wireless communication. The aircraft has a Lithium polymer battery which has to be charged before operation. The waypoints can be loaded prior to the mission or during flight. The aircraft is then mounted on the launch pad and is launched automatically. The aircraft has multiple modes of operation which allows the operator to switch to manual mode, semi-automatic mode or fully automatic mode. Once the mission is executed the aircraft can be directed to return back to launch or guided manually by the operator to the GCS.

IV. DESIGN CONSIDERATIONS

The entire UAV design cycle has been broken into three major phases as mentioned below to provide in a good better view for the understanding.

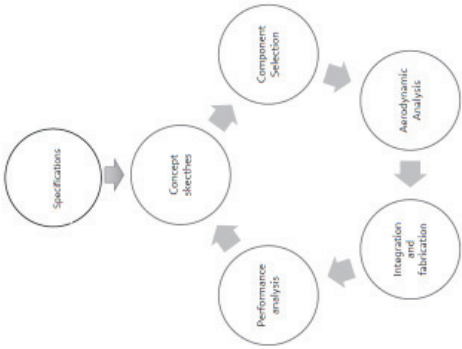


Fig. 1 Flow diagram of the design method.

A. Conceptual Design

The flow chart for the conceptual design process is given in figure below. The design requirements should come from the mission specifications. It includes parameters like range, operating altitude, endurance, weight, stability requirements, structural requirements etc. Then we need to analyze the availability of technology to meet the design requirements. It includes the availability of avionics, actuators, material for airframe and propulsion system. Cost factor also will decide the technology to be adopted for the design. The conceptual sketch indicates approximate wing and tail geometries, fuselage shape, location of avionics, actuators, propulsion system etc. This can be used to estimate aerodynamics and weight fraction by comparing with previous designs. Using these estimates, a first guess of weight, wing area and thrust requirement should be made. After this initial sizing of the UAV, a 3D drawing should be made with fuselage, control surfaces, and other components size and location indicated. Then initial aerodynamics, structural and thrust calculations to be done followed by tail geometry selection and component selection for avionics and actuators. After in initial analysis, the obtained performance and capabilities should be compared with the design requirements. Optimization methods are used for obtaining the best configuration based on criteria like light weight, higher endurance etc. If the design requirements are not met then an iterative process will continue until they are met. A revised layout is to be made including all the changes made in the optimization process. Then a detailed analysis of aerodynamics, materials and structural properties, propulsion, longitudinal and lateral stability and control, cost and fabrication methods to be done.

be used, holes to be drilled etc. Then the whole vehicle is broken into individual parts like wing, fuselage, vertical tail, control surfaces, motor mount and each part is analyzed separately. A major part of the detailed design is called production design. It explains about how each part will be fabricated, necessary procedures to be adopted and facilities and tools to be used for that. It starts with the smallest sub assembly like motor mount to the whole vehicle fabrication and assembly. Hardware in the loop simulation is done for testing actuators, control surfaces, control algorithm, autopilot hardware and software. After all this process, the final tested model is put into computer aided manufacturing machinery for production. This is accompanied by a detailed documentation and cost budgeting.

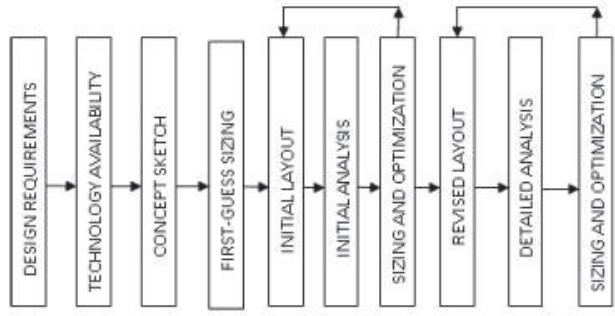


Fig. 2 Flow chart of conceptual design phase

This process will almost fix the control surface sizing, propeller sizing etc. Then this model has to be optimized for better performance. The output of the conceptual design phase will go through the preliminary design process.

B. Preliminary Design

Preliminary design begins when there are not much major changes to be made in the configuration. It starts with a 3D CAD model of the UAV with all the components included in appropriate locations. This CAD model has to be imported to other softwares for detailed CFD and structural analysis. Preliminary load calculation and stress analysis to be done including the thickness and location of the spar to be used. Structural testing should be done for determining the fatigue and failure point. Static and dynamic derivatives are to be derived and a full 6DOF non-linear model is to be developed. Then a detailed linear and non-linear simulation to be done to analyze vehicle handling qualities. Prototypes are built and tested in remote control mode first to determine the actual flying qualities of the vehicle. Then a wind tunnel model is made and tested to obtain the refined aerodynamic coefficients.

C. Detailed Design

Detailed design phase begins in which the actual pieces to be fabricated are designed. It starts with a complete 3D drawing including fine details like dimension of screws to

V. DESCRIPTION OF THE UAV

As you can see in Fig.3, it's in the normal extended flight mode now. In the design, there are two movable wings at the end of which are connected the brushless motor and the tri blade. These wings are connected to the high torque servo motors which are installed in the hull of the drone. These wings stay vertical during the take-off or landing mode. Adjacent the point of brushless motor is there is another servo motor whose shaft is connected to the secondary wing which unfolds during the flight mode. These secondary wings help for gliding in the air thus saving the power of the battery. Now to the back of the drone, we have the booster brushless motor which gives the speed to the UAV. They are also movable to a resolution of 90 degrees. During VTOL mode Fig.4, it's also in the vertical position, and this feature of having it vertical brings to the fact that it won't restrict the direction of take-off. This will cut down the wind and pause and instability coming in its way.



Fig. 3 Flight Mode

Now during the flight mode the main wings rotate 90 degrees to horizontal position and the secondary wings also retract or extend which brings it to the flight mode. The booster motors also rotate to horizontal positions and applying the necessary speed to the glider.



Fig. 4 Vertical Lift off and Landing Mode

This UAV as shown in Fig.5 is 2 meters long and provided with an elevator wing of the scale of .40 m. the profile of this wing is of type NACA 2412 with a cord of 0.236 m, stalled on the fuselage of 3° with a dihedral of 2°. The elevator group of the shape of T reversed is profiled with NACA 0012 and attached to the wings by two beams which give also the name of an UAV to tristar. These UAV's aspire of a remarkable stability to the no desirable movements of roll. This UAV is designed to have a booster motors which will be embarked behind the fuselage.

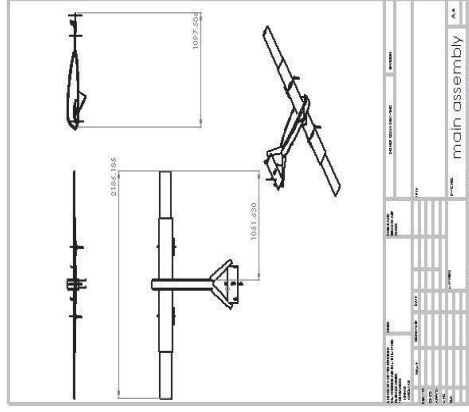


Fig. 5 2D CAD Model

VI. AERODYNAMIC DESIGN

The project aims at designing and building of a Class of portable and easily operable (by non-experts) Unmanned Air Vehicle (UAV) of fixed wing configurations. Though the UAVs can be used for diverse applications, this proposal aims at developing the UAVs for surveillance missions, where, for example, a man in motion can be identified from the aerial images displayed at the Ground Control station. UAVs are designed to work in the outdoor missions under light wind and showers (rain). A reliable robust digital integrated NGC (Navigation, Guidance and Control) system is integrated into the autopilot system of

the UAV. The various parameters involved in the mission are Range: It is the maximum distance that the UAV can travel away from its home station along any direction with normal mission payload, carry out its intended mission within the prescribed operating conditions, and return to home without reenergizing its propulsion system. Operating altitude: It is the altitude at where the specified payload performance (image quality) and video and data transmission is possible. Higher altitude will provide more coverage area and line of sight (LOS) for telemetry operations. But for focusing a particular region more, low altitude flight is desirable where there are not much tall buildings for hindrance. The UAV should be able to fly as low so that it can travel around trees.



Fig. 6 The final designed model of TSF

Loitering is an important feature of the UAV mission where it can continuously monitor a specific location. Loitering velocity and radius should be as low as possible for a better surveillance of a specific location. Here the minimum loiter radius is 15m at a velocity of 12 m/s. Endurance: It is the total time interval between UAV takeoff and landing. The endurance of the vehicle is a function of the total power available and the total power required. A high lift to drag ratio (L/D) of the wing results into higher endurance. The main factors affecting the L/D are the wing planform and the airfoil. The desired endurance for this vehicle is a minimum of 30 minutes. When the desired surveillance region is far from the home station, the UAV will climb to a certain height and will follow a leveled flight with minimum required number of turn till it reaches near the destination. After reaching near to destination location, it will be performing maneuvers like Climb, descent, turn or a combination of these. Maneuvering consumes more power when compared to straight and level flight, so the endurance depends upon the maximum Climb rate and minimum turn radius. Endurance of 60 minutes will be sufficient to navigate through a region less than 5 km of radius from the home station.

Total weight: The total weight of the vehicle is desired to be low in order to achieve low stall speed and make it easy to hand launch. This design parameter is affected by the choice of onboard equipment which in turn depends on the desired capabilities of the vehicle. The overall aerodynamic configuration of the vehicle contributes to this parameter. Another important factor affecting it is the construction material used. The desired total weight is about 4-6kg.

Thrust Available: The thrust available is a function of the size of the motor, desired thrust and the efficiency of the motor, battery and propeller combination. The thrust

margin on the aircraft is an indicator of the capability of the vehicle to take-off smoothly in presence of the launch speed variations. The required thrust available decides the motor size and weight. This in turn aspects the total weight of the vehicle. It is desirable to have at least 30% thrust margin.

Static Stability: The aircraft should be stable both longitudinally and laterally. The longitudinal static margin should be about 5% to 10%. This is desirable for good handling qualities. The static margin is dependent on the wing planform and airfoil used (deciding the center of pressure). It is also dependent on the location of center of gravity, which is in turn dependent on the fuselage design and component placement.

Structure robustness: The structure should be robust in order to sustain crashes in the initial flights. The repair time of the vehicle should be low.

TABLE 1
UAV DESIGN SPECIFICATIONS

Application :	Video surveillance
Payload :	Camera (8MPixel)
Take-off weight :	4Kg to 6 Kg
Size :	2m wing span
Normal flight speed :	10 - 25 m/sec
Maximum altitude :	200 meters (above ground)
Propulsion :	BLDC Motor with Unmanned propellers
Actuators :	Electro-mechanical
Power source :	Lithium Polymer cell
Flight Duration :	30 - 90 Minutes
Operation :	Clear/ rain/ dust, nil to moderate wind, Day time
Range :	50 Km
Climb rate :	Less than 5 m/s

UAVs are small aircrafts having cruising speeds of 10 to 25 m/s. Their cruising speeds correspond to a chord Reynolds number less than 2, 00, 000, and that has not been thoroughly studied. Flows over airfoils at low Reynolds numbers involve laminar, transitional, and turbulent boundary layers, sometimes with extensive regions of separated flow that can cover more than 50% of the chord. The low speed of the fluid leads to a low resistance to adverse pressure and free stream disturbances, which push the flow to an early separation. However, the laminar boundary layer yields an increased stability at low Reynolds numbers, resulting in early separation of the flow without prior transition to a turbulent regime. The flow may then reattach in a turbulent form called a laminar separation bubble. The formation of separation bubble depends largely on the geometry of the wing. The flow inside the bubble can induce transition to a turbulent flow. Such flow phenomena may appear at Reynolds numbers greater than 50,000. Gad-el-Hak [2001] showed that for Re less than 50,000, the separated flow will not reattach and the airfoil will stall at low to moderate angles of attack. For $50,000 < Re < 3,00,000$ the separated flow forms a laminar separation bubble. This reduces the aerodynamic efficiency substantially.

So the selection of airfoil and planform geometry is very critical for good aerodynamics.

A. Planform Selection

Aspect ratio was the most important factor for this as the aerodynamic performance is dependent by wing planform and operating Reynold's number. To cover in the maximum area and to get the most of the aspect ratio, the rectangular planform was selected giving in the maximum lift curve. We get the maximum area with rectangular than the Zimmerman and circular planforms giving a better lift ratio for the weight concerned. We see that an increase in aspect ratio will increase the slope of the alpha versus co-efficient of the lift curve and will reduce the induced drag. This will improve C_L/C_D ratio. With the flight test results, rectangular planform has been giving the maximum efficiency for aerodynamic performance and we chose it to be the best for our purpose.



Fig. 7 Front View of Gliding mode TSF

B. Airfoil Selection

TABLE 2
UAV PARAMETERS

Planform	Rectangular
Airfoil Modified	NACA2412
Weight (grams)	3-6Kg
Velocity(m/s)	10-25
Chord(m)	0.2
Span (m)	2.0

The moment of inertia values are obtained from SOLIDWORKS.

C. Aerodynamic Modeling using XFRLR

The first step towards aerodynamic modeling is to make a model of our UAV in XFRLR which is shown below.



Fig. 8 Model of TSF used in XFRLR

A batch analysis is carried out by varying the velocity, angle of attack and elevator deflection. The angle of attack is varied from 0 degrees to 15 degrees and analyzed. The velocity is varied from 10m/s to 25m/s. Note that only one

vertical fin can be added at a time for the simulation to be processed. After simulating individually, the set of data is converged to get a common set of data for the entire model. As XFRLR gives huge set of data at each operating point, this data is processed using MATLAB. Specifically curve fit is done to obtain the aerodynamic coefficients as a function of velocity, angle of attack, elevation deflection.



Fig. 9 Pressure distribution on Wing (Cp)

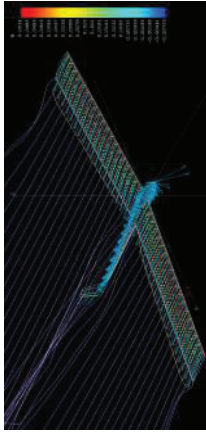


Fig. 10 Lift force distribution along with Flow visualization

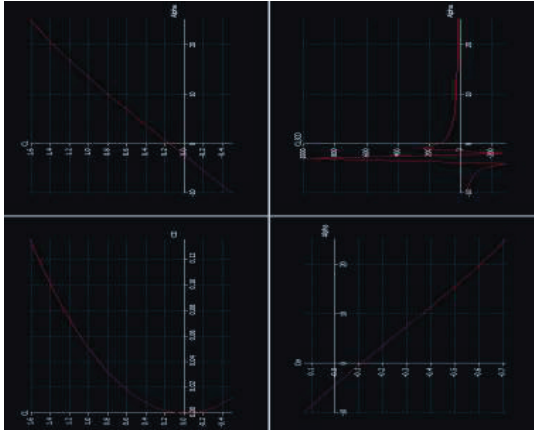


Fig. 11 Cl & Cd Graphs of the XFRLR Model

D. Airfoil Wind Tunnel Test

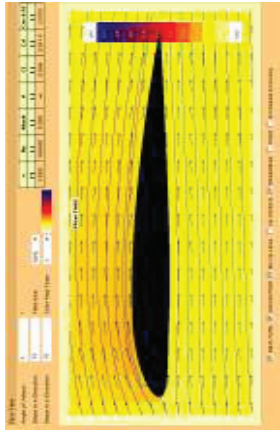


Fig. 12 Velocity Ratio Analysis

Several compilations were carried out for various angles of incidence to determine the lift and drag. Each case was carried out at the same speed: 10 m/s. The angle of attack extended from 3 to 18 degrees. It is a parabolic type tendency similar to those found on the planes in general. In particular, one notes the increase in the drag with the increase in the angle of incidence. This curve is presented with drag of the isolated wing which equips the UAV in question and which was obtained by wind tunnel test. The two diagrams follow the same tendency separately which is normal because variation of the drag between the two curves corresponds to the drag of the remainder of the UAV which varies very slightly with the incidence. The current NACA airfoil proves to be having a low turbulence and a better lift factor with the angle of attack set at 3degrees. The velocity ratio is balanced properly. And the pressure generation around the airfoil is distributed symmetrically converging into a stable airfoil design.

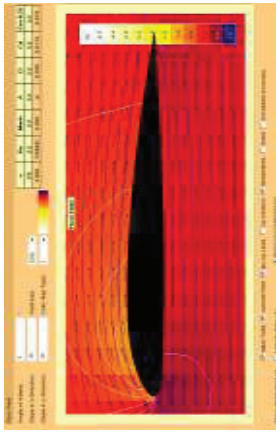


Fig. 13 Pressure Co-efficient Analysis

E. Eppler & NACA Analysis

An agreement of the two curves is noted in the linear zone which is explained by the fact that it is the wing which contributes in a great proportion to the total lift produced on the plane. The other parts of the plane, in particular the stabilizer contribute slightly towards weak and moderate

incidence. However the lift of the isolated wing exceeds

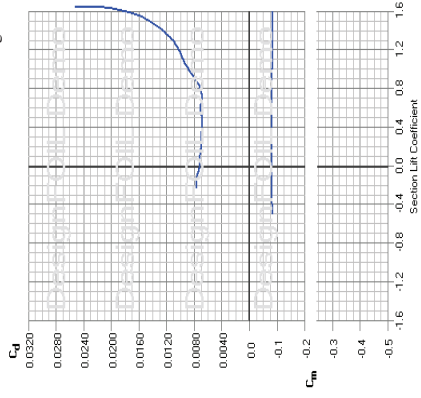


Fig. 14 NACA Analysis of Cd vs Cl

that of the plane for the bigger angles of incidence. This is due with the fact that the Reynolds number of the tests of the isolated wing is much higher than that of the plane tested numerically which leads to higher maximum C_m for the isolated wing. Numerical calculations show well the phenomenon of unhooking of the UAV which is located at approximately 3 degrees. It should be noted that this incidence which appears is weak only seemingly because it should be mentioned that the reference of measurement of the angle of attack is the base of the fuselage and that the wing is fixed with 4 degrees. flow return is noted on almost the totality of the wing. in particular, it is much attenuated in the central part.

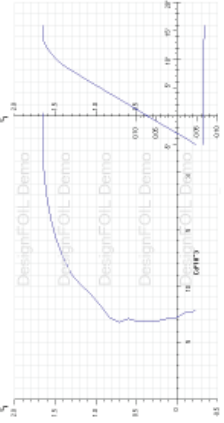


Fig. 15 Eppler Analysis of Cd vs Cl

This effect is expected because the wing is not twisted and the great deflections of the air are perceptible in this area. The separation of the flow recorded on the nose of the fuselage still persists with this angle of incidence, flow recorded on the level of the wing root with the fuselage with 15° of incidence.

F. Solidworks 3D Modelling

A slight modification was made to the pre-existing design by adding a motor mount and to add components like Battery, ESC, Autopilot etc. The CAD model of the final design was made in Solidworks. Fig.16 shows the model. The proper modeling of each of the components

were done so as to find the exact CG location and Moments to get the proper plant model of the UAV. The exact airfoil of the UAV is imported to get the further flow analysis over the surface with proper flow separation due to surface friction even.



Fig. 16 Solidworks designed model

G. CFD Analysis of Fuselage & Airfoil

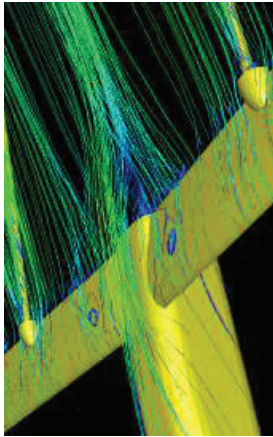


Fig.17 Air flow around Fuselage & Airfoil

The flow simulation done with the propeller effects give in a perpetual concept for the particular flow pattern given by the particular propeller and the orientation of the motor mount. The simulation done below gives in a basic idea with a complete horizontal mounted motor with and a 4inch propeller. The flow tends to be uniform and gives good results.

It is a representation of the streamlines of the flow inside the internal zone of the boundary layer around the UAV which is completely pressure building. It is a very useful diagram for the draftsman of the aerodynamic forms because it reveals the design defects characterized by the appearance of the zones of flow separated color blue as shown in the Fig.17 on the wing and on the level from the bases from fastener from the wings. Fig. 18 shows the two marginal swirls which release from salmons of the two wings of the UAV to the passage of the flow for the incidence 15°, the size of the marginal swirl for such an incidence is considered to be considerable. Moreover, one notes the complexity of the flow around the plane for an incidence judged like the beginning of turbulence.

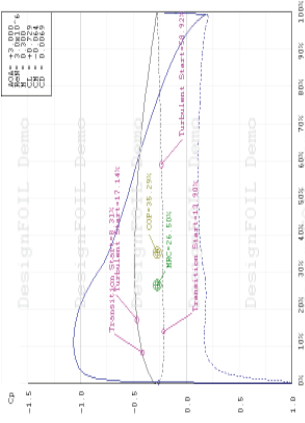


Fig.18 Turbulent Analysis

The trailing edge leads to a better flow with a slight generation of turbulence at the top camber line which is obvious as there generation of pressure built at that position.

H. Wing Stress & Loading Analysis

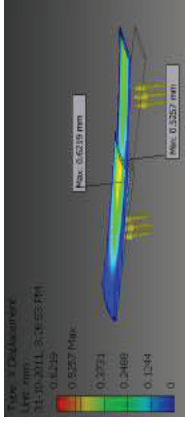


Fig.19 Wing Loading Effect

At this time, the maximum load will be acting upon the wing region and hence analyzing the wing for structural stability will be enough to ascertain the stability of the total model. Hence finite element model of the wing region along with the mount was made. The CAD model thereof was exported to the Ansys Workbench Geometric Modeler. The contacts between wing section and the mount were treated as bonded contacts. Meshing was done to the model with a global medium relevance center with automatic size feature ON for curvature. Also, fine meshes were given for all the contacts and regions of probable failure. Fig. 19 shows the Equivalent stress (Von-Mises) distribution in the wing when subjected to the loading conditions discussed before.

While both aerodynamic and structural metrics are included in the wing design, the analysis are not explicitly coupled in an aerodynamic sensor, stable deflections of the wing due to air loads are assumed to be small, since large ones will lead to buckling failure. Aerodynamic performance of the wing is studied using the fixed wing stress analysis. Aerodynamic analysis computes the pressure distribution over the wing, as well as the concomitant lift, lift-to-drag ratio (L/D) and the static stability margin. An arc length method based nonlinear deviation tool is used to evaluate the structural performance of the wing and to determine maximum flying velocity that the wing can withstand

without buckling or failing (laminate failure with factor of safety = 1.5) under aggressive flight loads.

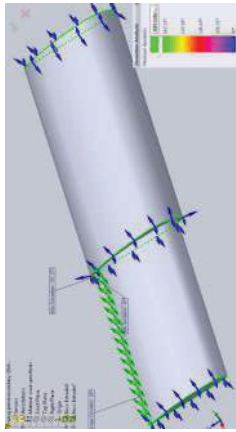


Fig. 20 Points of Loading

Experimental validation of the numerical procedure used for determining maximum flying velocity has been carried out and can be found elsewhere. An analytical approach based on classical laminate plate theory and strain superposition (initial chord-wise strains due to root airfoil flattening and additional span-wise strains due to incremental rolling) is used to study the stresses developed in the composite wing during the rolling process for storage with a safety factor of 1.5 is used to check first ply failure of the composite wing to determine minimum safe storage diameter.

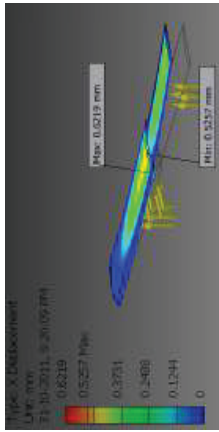


Fig. 21 Thrust & Loading Effect

Taking advantage of problem symmetry, we model only one wing half, with the appropriate symmetric boundary conditions at the root airfoil. And appropriate results have been found as the stress analysis proved correct and the wing was able to sustain the pressure at the trailing edge as we rarely see reddened part on the airfoil. Reaching to a corrected figured value achievable to 10m/s or 20m/s.

I. UAV Curvature & Smoothness Analysis

In the deterministic optimization study, wing airfoil shape, wing taper ratio, wing geometric twist angle, and wing quarter chord sweep angle are considered as design variables, as well as the lay-up scheme of the composite bendable wing. The airfoil at the root of the wing is completely defined by four variables, the camber towards the leading edge, the chord-wise position of this control point, and similar information for the camber towards the trailing edge. These variables are defined to be the chord normalized values.



Fig. 22 VTOL Mode Curvature

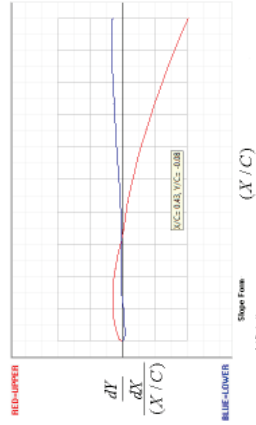


Fig.23 First Derivative Curve

A quadratic curve from the leading edge to the maximum camber point, which takes in the first derivative, a polynomial in the middle and another quadratic curve from reflex point to the trailing edge, can completely and uniquely define root airfoil shape and its smoothness. Curvature continuity is ensured between the color combinations. For a constant camber wing design the tip airfoil geometry is the similar as the root airfoil, while for a variable camber wing design the tip airfoil is defined independent of the root airfoil.



Fig. 24 Flight Mode Curvature

All the metrics are interpolated between the root and the tip to obtain a kink free wing shape, as can be graphically seen above.

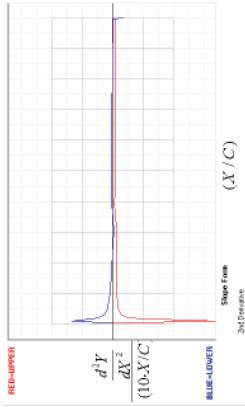


Fig. 25 Second Derivative Curve

Due to manufacturing constraints (plain weave bi-directional nature of material used in the manufacturing) and to ensure symmetric deformation of the two wing halves under aerodynamic loading, only symmetrized layout orientations are allowed.

VII. SYSTEM ARCHITECTURE

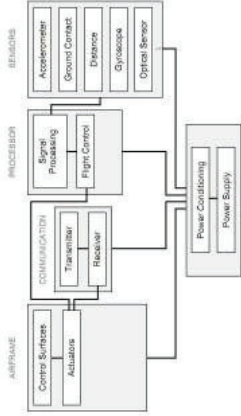


Fig. 26 Architecture

This section gives an overall view of the system architecture with all the sensors, dynamics and feedback control system. Fig. 26 describes the basic architecture of the model with all the subsystems. Fig. 27 shows the state flow of the command and data flow to the controller and how the actuators. This loop shows the logic mechanics and how its implemented for the entire model. Fig. 28 shows the flight software architecture which decides the flow of data and its prioritizing for different set of commands.

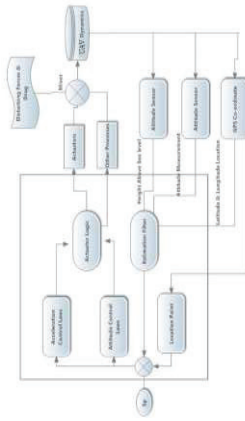


Fig. 27 State Flow Diagram

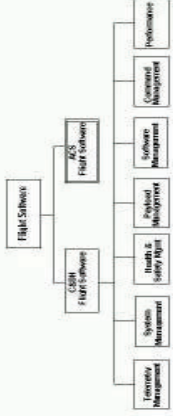


Fig. 28 Flight Software Architecture

VIII. TSF KINEMATICS & DYNAMICS

A. Mathematical Modeling

Mathematical modeling provides a description of the behavior of a system. The flight behavior of a triopter is determined by the speeds of each of the four motors, as they vary in concert, or in opposition with each other. Hence, based on its inputs, a mathematical representation of the system can be used to predict the position and orientation of the triopter. The same can further be used to develop a control strategy, whereby manipulating the speeds of individual motors results in achieving the desired motion to derive the full mathematical model of the triopter. We need to define its kinematics. And dynamics first, the kinematic equations provide a relation between the vehicle's position and velocity, whereas the dynamic model defines the relation governing the applied forces and the resulting accelerations

B. Reference frames

Before getting into the equations of kinematics and dynamics of the triopter, it is necessary to specify the adopted coordinate systems and frames of reference, as well as how transformations between the different coordinate systems are carried out. The use of different coordinate frames is essential for identifying the location and attitude of the triopter in six degrees of freedom (6 DOF). For example, in order to evaluate the equations of motion, a coordinate frame attached to the triopter is required. However, the forces and moments acting on the triopter, along with the inertial measurement unit (IMU) sensor values, are evaluated with reference to the body frame. Finally, the position and speed of the triopter are evaluated using GPS measurements with respect to an inertial frame located at the base station.

Thus, three main frames of reference are adopted, as shown in Figure 29:

1. The inertial frame, $Fi = (\bar{X}_i, \bar{Y}_i, \bar{Z}_i)$, is an earth-fixed coordinate system with the origin located on the ground, for example, at the base station. By convention, the x-axis points towards the north, the y-axis points towards the east, and the z-axis points towards the center of the earth.
2. The body frame, $Fb = (\bar{X}_b, \bar{Y}_b, \bar{Z}_b)$, with its origin located at the center of gravity (COG) of the triopter, and its axes aligned with the triopter structure such that the x-

axis $\bar{X}_b$ is along the arm with front motor, the y-axis $\bar{Y}_b$ is along the arm with right motor, and the z-axis $\bar{Z}_b = \bar{Y}_b \times \bar{X}_b$, where 'x' denotes the cross product.

3. The vehicle frame, $Fv = (\bar{X}_v, \bar{Y}_v, \bar{Z}_v)$, is the inertial frame with the origin located at the COG of the triopter. The vehicle frame has two variations, Fv and $F\theta$. Fv is the vehicle frame, Fv , rotated about its z-axis $\bar{Z}_v$ by an angle ψ so that $\bar{X}_v$ and $\bar{Y}_v$ are aligned with $\bar{X}_b$ and $\bar{Y}_b$, respectively. $F\theta$ is frame Fv rotated about its y-axis, $\bar{Y}_\theta$, by a pitching angle, θ , such that $\bar{X}_\theta$ and $\bar{Z}_\theta$ are aligned with $\bar{X}_b$ and $\bar{Z}_b$, respectively.

Translation and rotation matrices are used to transform one coordinate reference frame into another desired frame of reference. For example, the transformation from Fi to Fv provides the displacement vector from the origin of the inertial frame to the center of gravity (COG) of the triopter. Also, the transformation from Fv to Fb is rotational in nature, therefore yielding the roll, pitch and yaw angles.

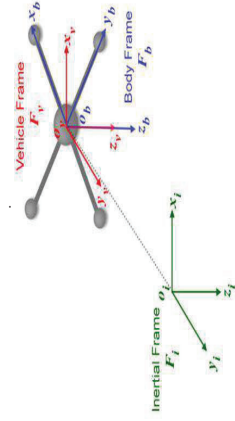


Fig.29 Inertial, body and vehicle frames of reference.

C. Triopter's kinematics

Let $\vec{p}_F^T = [Px, Py, -Pz]$ and $\vec{\Omega}_F^T = [\dot{\theta}, \dot{\psi}]$ denote the triopter's position and orientation within a given frame F. The relation between the triopter's speed in the three predefined frames is expressed as:

$$\begin{bmatrix} \dot{P}_x \\ \dot{P}_y \\ -\dot{P}_z \end{bmatrix}_{Fi} = \begin{bmatrix} R_{Fb}^T & \dot{P}_x \\ \dot{P}_y \\ \dot{P}_z \end{bmatrix}_{Fb} \quad (1)$$

Where

$[R_{Fb}^T] \in R^{3 \times 3}$ is the rotation matrix that maps frame Fb to frame Fi and is defined by

$$[R_{Fb}^T] = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

with $c\theta = \sin\theta$ and $c\theta = \cos\theta$. The same notation applies for $s\varphi$, $c\varphi$, $s\psi$, and $c\psi$. The rotational motion relationship can therefore be derived using the appropriate state variables, such as the vehicle frame angles (φ , θ , and ψ) and the body frame angular rate $\dot{\theta}$. However, in order to do so, these variables need to be brought into one common frame of reference. Using rotation matrices to transform vehicle frames F_θ , F_θ , and F_ψ into the body frame of reference F_b , we get :

$$\begin{bmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}_{F_b} = R_{F_b}^{F_\theta}(\theta) \begin{bmatrix} \dot{\varphi} \\ 0 \\ 0 \end{bmatrix} + R_{F_b}^{F_\theta}(\theta) R_{F_\theta}^{F_\psi}(\psi) \begin{bmatrix} \dot{\theta} \\ 0 \\ 0 \end{bmatrix} + R_{F_b}^{F_\psi}(\psi) R_{F_\psi}^{F_\psi}(\psi) \begin{bmatrix} \dot{\psi} \\ 0 \\ 0 \end{bmatrix} \quad (3)$$

Where

$$R_{F_b}^{F_\theta} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\theta & s\theta \\ 0 & -s\theta & c\theta \end{bmatrix}, \quad R_{F_\theta}^{F_\psi} = \begin{bmatrix} c\theta & 0 & -s\theta \\ 0 & 1 & 0 \\ s\theta & 0 & c\theta \end{bmatrix}$$

and

$$R_{F_b}^{F_\psi}(\theta) = R_{F_b}^{F_\theta}(\theta) R_{F_\theta}^{F_\psi}(\psi) = 1 \quad (4)$$

Therefore,

$$\begin{bmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}_{F_b} = \begin{bmatrix} 1 & 0 & -s\theta \\ 0 & c\theta & s\theta c\theta \\ 0 & -s\theta & c\theta c\theta \end{bmatrix} \begin{bmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}_{F_\psi} \quad (5)$$

IX. DYNAMIC MODEL OF THE UAV

The vehicle is equipped with 2 wings that are mounted on the front and an elevator at back of the vehicle and can be rotated from vertical to horizontal positions. Figure 30 below shows the aerial vehicle in vertical flight modes.

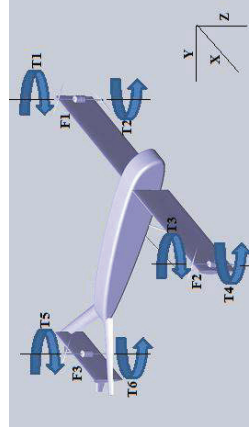


Fig.30 Aerial Vehicle in Vertical Flight Modes

With this wing configuration, the vehicle's airframe transforms into a triopter like structure if the wings are at

the vertical position and into an airplane like structure if the wings are at the horizontal position. To keep the control complexity on a minimum level, the rotations of the wings are used as attitude control inputs in addition to motor thrust inputs of the vehicle in horizontal flight mode. Therefore the need of additional control surfaces that are put on trailing edges of the wings on a regular airplane are eliminated. Two wings on the front can be rotated independently to act like the ailerons while wing at the back are rotated together to act like the elevator. This way the control surfaces of a regular plane in horizontal flight mode are mimicked with minimum number of actuators.

A. Vertical Flight Mode

In vertical flight mode, the mechanical structure of the vehicle is very similar to a triopter vehicle and the dynamics of the vehicle can be approximated by the model given below: Fig. 15 Aerial Vehicle in Vertical Flight Configuration. The two reference frames given in Fig. 15 are body fixed reference frame B : (Ob,xb,yb,zb) and earth fixed inertial reference frame W : (O,x,y,z). Using this model, the equations describing the position and attitude of the vehicle are obtained by relating the 6 DOF kinematic equations with the dynamic equations. The position and linear velocity of the vehicle's center of mass in world frame are described as,

$$\begin{bmatrix} \dot{X} \\ \dot{Y} \\ \dot{Z} \end{bmatrix} = \begin{bmatrix} \dot{X} \\ \dot{Y} \\ \dot{Z} \end{bmatrix} = \begin{bmatrix} \dot{X} \\ \dot{Y} \\ \dot{Z} \end{bmatrix} \quad (6)$$

The attitude and angular velocity of the vehicle in world frame are given as,

$$\alpha = \begin{bmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}, \quad \omega = \begin{bmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} \quad (7)$$

where, φ , θ , ψ are named roll, pitch and yaw angles respectively. The equations for the transformation of the angular and linear velocities between world and body frames are given in equations (1) and (2):

$$\begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{bmatrix} = T_{bw}(\varphi, \theta, \psi) \begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{bmatrix} \quad (8)$$

Where

$$T_{bw}(\varphi, \theta, \psi) = R_x(\varphi) R_y(\theta) R_z(\psi) \quad (9)$$

$$W_b = \begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = E(\varphi, \theta) \cdot \omega \quad (10)$$

Where

$$E(\varphi, \theta) \cdot \omega = \begin{bmatrix} 1 & 0 & -\sin(\theta) \\ 0 & \cos(\theta) & \sin(\theta) \cos(\theta) \\ 0 & -\sin(\theta) & \cos(\theta) \cos(\theta) \end{bmatrix} \begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix}$$

The dynamic equations for the forces and torques acting on the vehicle's center of gravity are given in equations (3) and (4):

$$F_{tot}^v = M_b \ddot{V}_b + \omega_b \times (M_b \cdot V_b) \quad (11)$$

Where

$$M_b = \begin{bmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & m \end{bmatrix}$$

$$M_{tot}^v = I_n \dot{\omega}_b + \omega_b \times (I_n \cdot \omega_b) \quad (12)$$

Where

$$I_n = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}$$

I_{xx} , I_{yy} and I_{zz} are moment of inertias in the body reference frame. The total force F_v tot acting on the vehicle's center of gravity is the sum of the lifting forces F_v eg created by the rotors, the gravity F_g and the aerodynamic forces F_a which is considered as a disturbance, namely

$$F_{tot}^v = F_{eg}^v + F_g + F_a^v \quad (13)$$

Where

$$\begin{bmatrix} F_{eg}^v \\ F_g^v \\ F_a^v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$F_g = \begin{bmatrix} -\sin(\theta) \cos(\theta) \\ \cos(\theta) \cos(\theta) \\ \sin(\theta) \end{bmatrix} \cdot mg$$

The total torque M_v tot acting on the vehicle's center of gravity (cog) is the sum of the torques M_v eg created by the rotors and the aerodynamic torques M_a which is again considered as a disturbance, namely

$$M_{tot}^v = M_{eg}^v + F_a^v \quad (14)$$

Where

$$M_{eg}^v = \begin{bmatrix} M_{xx}^v \\ M_{yy}^v \\ M_{zz}^v \end{bmatrix} = \begin{bmatrix} l_s & -l_s & l_s \\ l_l & l_l & -l_l \\ \lambda_1 & \lambda_2 & \lambda_3 \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix}$$

l_l and l_s are the distances of rotors to the center of gravity of the vehicle in x and y directions. Note also that $T_i = \dot{\varphi}_i$ for $i = 1, 2, 3$ and sum of torques created by the rotors result in a yaw moment along the z axis in vertical flight mode:

$$M_z^v = \sum_i T_i \quad (15)$$

B. Horizontal Flight Mode

In horizontal flight mode, the mechanical structure of the aerial vehicle forms the airframe of a four winged airplane with two wings on the front and two wings at the back. The

aerodynamic lift and drag forces that are considered as external disturbance in vertical flight mode need to be taken into account in dynamic modeling for horizontal flight. Fig. 16 shows the aerial vehicle in horizontal flight mode. The lift forces F_{il} and drag forces F_{id} for $i = 1, 2, 3$ generated by each wing are obtained from the equations:

$$F_{il}^i = -\frac{1}{2} C_{L_i}(\theta_i) \rho A_i V^2 \quad (16)$$

$$F_{id}^i = -\frac{1}{2} C_{D_i}(\theta_i) \rho A_i V^2 \quad (17)$$

where the angle of attack for each wing is given by q_i . Note that the lift coefficient $C_L(q_i)$ and the drag coefficient $C_D(q_i)$ are not constant variables, but functions of angle of attack q_i for each wing. The total force F_h tot acting on the vehicle's center of gravity is the sum of the forces F_h eg created by the rotors, the gravity F_g , the lift and drag forces generated by the wings.

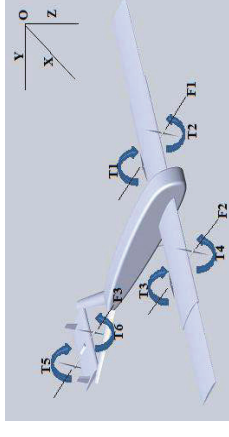


Fig. 31 Aerial Vehicle in Horizontal Flight Modes

F_w and the aerodynamic forces F_h a which is considered as a disturbance, namely

$$F_{tot}^h = F_{eg}^h + F_g + F_w + F_a^h \quad (18)$$

Where

$$F_{eg}^h = \begin{bmatrix} F_{x1}^h \\ F_{y1}^h \\ F_{z1}^h \end{bmatrix} = \begin{bmatrix} F_1 + F_2 + F_3 \\ 0 \\ 0 \end{bmatrix}$$

$$F_w = \sum_i F_{il}^i + \sum_i F_{id}^i = \begin{bmatrix} -(C_{D_1}(\theta_1) + C_{D_2}(\theta_2) + 2C_{D_3}(\theta_3)) \frac{\rho A}{2} V^2 \\ 0 \\ -(C_{L_1}(\theta_1) + C_{L_2}(\theta_2) + 2C_{L_3}(\theta_3)) \frac{\rho A}{2} V^2 \end{bmatrix}$$

The total torque M^h tot acting on the vehicle's center of gravity is the sum of the torques M_h eg created by the rotors, M_w created by the drag/lift forces of the wings and the aerodynamic torques M_a which is again considered as a disturbance, namely

$$M_{tot}^h = M_{eg}^h + M_w^h + M_a^h \quad (19)$$

Where

$$M_{CG}^h = \begin{bmatrix} M_x^h \\ M_y^h \\ M_z^h \end{bmatrix} = \begin{bmatrix} -\lambda_1 & -\lambda_2 & -\lambda_3 \\ 0 & 0 & 0 \\ I_x & -I_y & I_z \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix}$$

$$M_w = \begin{bmatrix} -(C_L(\theta_1) - C_L(\theta_2)) \frac{\rho A}{2} V^2 l_s \\ -(C_L(\theta_1) + C_L(\theta_2) - 2C_L(\theta_3)) \frac{\rho A}{2} V^2 l_i \\ -(-C_D(\theta_1) + C_D(\theta_2)) \frac{\rho A}{2} V^2 l_s \end{bmatrix}$$

Note that the sum of torques created by the rotors result in a roll moment along the x axis in horizontal flight mode

$$M_x^h = \sum T_i \quad (20)$$

The dynamic equations obtained for 6 DOF rigid body transformation of the aerial vehicle in inertial reference frame W are also valid for the dynamic model in horizontal flight mode:

$$F_{tot}^h = M_b \ddot{V}_b + \omega_b \times (M_b \dot{V}_b) \quad (21)$$

$$M_{tot}^h = I_n \dot{\omega}_b + \omega_b \times (I_n \omega_b) \quad (22)$$

X. CONTROL LOOPS

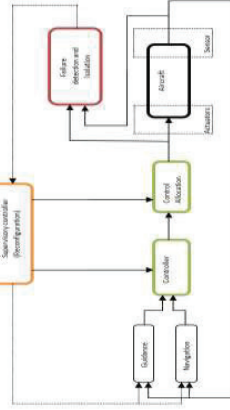


Fig. 32 Basic Control Loop Architecture

For TSF, a PID controller was chosen due to its proven capability in similar systems and its ease of implementation. Because the vehicle was a new design, it was immediately apparent that the time to develop, build, test, and refine the design of this vehicle would be substantial. Thus, an effective yet relatively simple control scheme was ideal, because it minimized the time required to solve this important and complex problem. A number of similar applications and vehicles which used a PID controller were found and it was discovered that the performance of this type of controller would be adequate to complete the competition objectives. Additionally, this type of controller allowed returning based on observable behavior. Once the gains were set near the optimal values it was possible to tune the vehicle empirically, which reduced

the reliance on the (potentially incomplete) dynamic model and provides a measure of robustness.

In any PID controller, the system input is comprised of three terms which are related to the difference (error) between the current system state and a desired state. The proportional (P) term multiplies the error by a gain value and uses this as an input. This increases the influence of this term when the current state is further away from the desired state. The integral (I) term uses the integral of the error over time multiplied by a gain value to influence the input. This generally has a gradual effect that reduces steady state error. The derivative (D) term, takes the derivative of the error and multiplies this by a gain value to influence the input based on how the error of the state is changing. This has a dampening effect which decreases oscillations in the system response. TSF use four PID controllers in each of its roll, pitch, yaw, and height maneuvers.

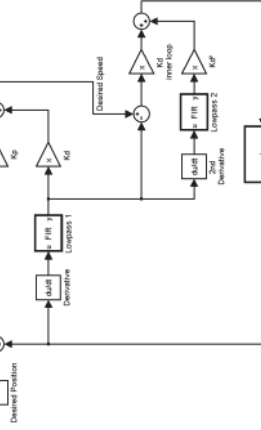


Fig. 33 Filtered Control structure

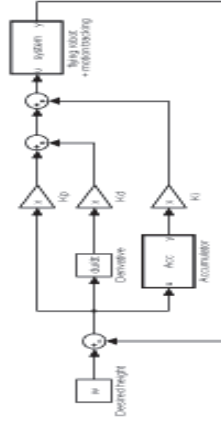


Fig. 34 Conditioning Loop

The onboard controllers are three independent PD loops, one for each rotational axes (roll, pitch and yaw). Angular velocities measured by the gyroscopes and computed relative angles are used as inputs. The angles are derived by integrating the sum of the output of one gyroscope and an external control input for the respective axis. Without an external input signal the calculated integral represents the angle the flying robot has turned in the respective axis. Looking at the closed loop and disregarding measurement noise and integration errors, this means that the robot will always keep its current orientation. The integrated angles

can be shifted by an external control input. As a result, the UAV's orientation changes.

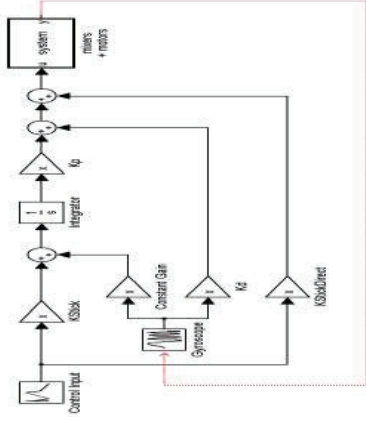


Fig. 35 PID loop

Proportionally to the input, its movements are controlled by steering it to a certain orientation and keeping this orientation for a certain time. Due to measurement noise and discrete integration the integrated angles drift about ± 3 degrees per minute. However, this drift can be easily compensated by a human pilot or an autonomous external position control such as a motion capture system. The controller implementations have been optimized for shortest possible execution time and robustness in almost every flight situation. The loop is interrupt triggered, which enables stable time constants for integration and filtering. By using the AVR's internal ADCs at a high sampling rate, fixpoint arithmetics only, runtime optimized FIR filter implementations and interrupt driven communication to update the motor speeds, we achieved a system running at a control frequency of 8MHz.

XI. SIMULATION RESULTS

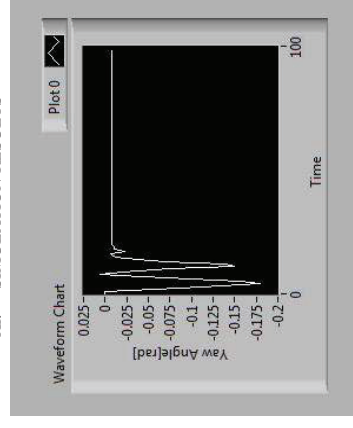


Fig. 36(a)

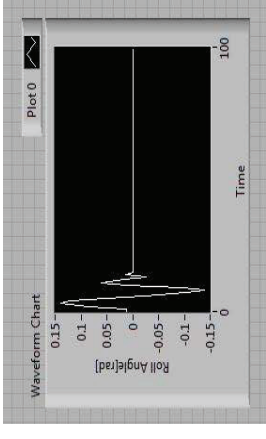


Fig. 36(b)

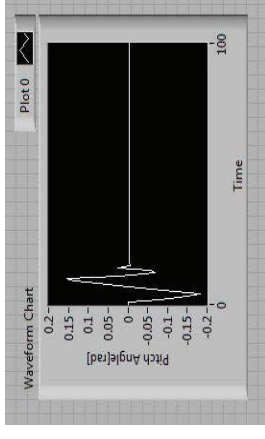


Fig. 36(c)

Fig. 36(a)(b)(c) Attitude control of the vehicle under no disturbance condition

Fig. 36 show the variation of the position and attitude state variables for the reference inputs $P_r = (3.3, -1.5)^\top$ and $y_r = 0$ under ideal operating conditions without any external disturbance. Note that position and angle references are tracked with negligible steady state errors.

XII. GROUND CONTROL STATION

TSF vehicle status can be monitored through the graphical user interface (GUI). The GUI reports critical flight data such as vehicle altitude, battery life, and communication signal strength as well as the mission state. With this information, we can predict what issues could possibly arise allowing us to better react to a situation and reduce risk as much as possible. Fig. 37 shows the loop flow for the ground control centre command flow.

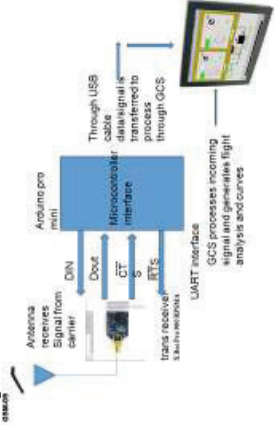


Fig. 37 GCS Flow Architecture

A. Material Selection

There are a variety of materials available for the construction of bigger size UAVs. Weight and strength being the primary concern, the possible material combinations are:

1. Depron is dense foam which is strong, sturdy and easy to work with. But it is slightly brittle and has a tendency to form dents.
2. Balsa is a very low density wood with very good strength to weight ratio. Balsa also has similar properties like Depron but is stronger. But the fact that it is a type of wood causes different batches to have different properties causing inconsistencies in its nature. It is also vulnerable to pest, atmospheric temperature, humidity causing it to warping and degradation to strength.
3. Carbon fiber is very strong and can form the frame work of the aircraft. A light weight skin could cover the rest of the surface to provide an aerodynamic surface.
4. EPP is lightweight, tough and very flexible and is a good choice for external structure. The UAV was modeled using SOLIDWORKS R with all its components. Mating and control section actions are checked for obstruction and optimal spacing is provided for all control surfaces. The Centre of Gravity of the UAV is appropriately positioned as per the static margin requirements. The detailed assembly and material used for each part are explained below.

Tail: Made of depron (density= $40\text{kg}\cdot\text{m}^{-3}$) as it is very light weight, can be replaced by balsa wood of 2mm thickness with a compromise that since balsa is very brittle, it might break at greater velocity or crash.

Elevator: The elevator profile is normal rectangular shape and is also made of depron and may be replaced by balsa.

Wing: The airfoil is the one which is been finalized after the thorough analysis done in XFRLR and being the best for the particular sized wing. The wing would be made of EPP as it maintains the proper airfoil shape throughout the entire span and provides enough shock absorbing properties.

Propeller: The required propeller specification is 5×4 inch which is sufficient enough for required thrust.

Rudder also has been provided and made of depron.

Electronic components: Battery used is of 6000mAh which gives enough endurance to the UAV. The PCB is fabricated at a higher level scale of 8 layers to reduce the size of the on board system. Electronic speed controller is separated from main PCB, which eliminates power transition effect and fluctuations.

B. Electrical Sub-Assembly



Fig. 40 EEPS Architecture

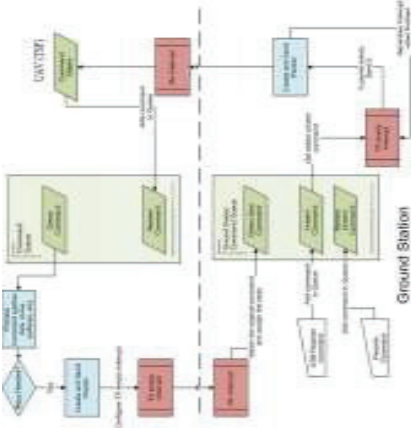


Fig. 38 Command Flow for GCS



Fig. 39 Ground Station GUI Demo

XIII. FABRICATION AND ELECTRONICS ASSEMBLY PLAN

The UAV features a modular architecture with following parts: Main Body, Wing with a Avionics bay, Tail section, Propulsion unit, Actuators, Control surfaces. Avionics (RC): Stabilizer, Receiver, Battery, Electronic speed control.

The components are connected, arranged and glue firmly in place using a jig on to the slider tray. The avionics panel has a jumper for charging, which is positioned on the bottom side of the avionics panel and will be easily accessible by the operator on the belly of the UAV. For charging the jumper can be removed and the charger can be plugged in at its place. The jumper is colored bright orange to avoid it being not plugged in before flight.

The model view of all the electronics and flow diagram is given below. Fig. 40 shows the all the electrical and electronics flow diagram. This helps in proper current allocation and digital component isolation from high EMF regions.

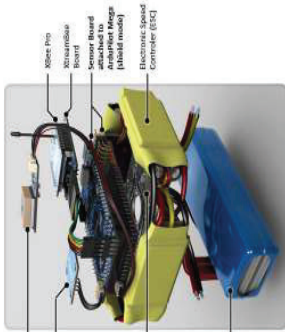


Fig. 41 Model View of EEPS



Fig. 42 Autoilot board for the tests

The Test Avionics unit size is 80 mm length, 40mm width and 10mm thickness and these constraints easily fit into the modified airfoil configuration.

The Avionics panel consists of the connectors for the actuators and the motor. Once these wires are connected to the main body, the avionics panel can be slide into the main body with the help of the carbon fiber guides positioned inside the main body.

XIV. CONCLUDING REMARKS

The present study was undertaken to observe the aerodynamic performances of an UAV by using a CFD computer code and to find out the system mathematical model. This concept of design of flying machines is judged like an alternative to the wind tunnel tests for a possible reduction of the design cost. This work brought moreover, certain important aerodynamic corrections around this UAV such as the aerodynamic stability of the nose of the fuselage and the setting of the bases of fastener of the beams on the level of the under-surface of the wings with

the proper stress analysis done and higher stability of the UAV under different situations.

Several change and simplification were made in the modeling process, including the use of electric brushless motors instead of reciprocating, the constant mass, counter-rotating propeller to eliminate rotating-mass effects, the removal of aileron control, etc. These changes were made in a way that the flight characteristics would not change much. The simulations were conducted several times performing a take-off, hover, cruise and landing maneuver. Software like Solidworks, Autodesk Inventor, Design Foil, Matlab, Simulink, Labview and many others have contributed a lot for the analysis and simulations done in this paper. It's been a successful path along

Obtaining a proper design procedure and following the method in the process mentioned above has given on good results. The flight has been flown with RC mode and data logging has been done with the Autoilot Board mounted which is helping us to design the algorithm and to trim the gains values effectively. The XFRLR data has provided the basic data to which the norms were set and design was preceded further to Solidworks modeling and FEA/Flow Simulations that was carried in Ansys. The results gave us good data as to which the final design was proposed. Various material and fabrication methods were tested and final configuration material was EPP as for its good shock absorbing capability and light weightness. After the final model fabrication, the advanced control algorithms will be tested to make it a robust autonomous flight for Tri-Star Falcon (TSP).

ACKNOWLEDGMENT

We would like to thank the college authorities, the faculty members and our guides Dr. V. Juliet, Dr. S.N. Pillai, Asst.Prof. V. Prabhu, Asst. Prof. G. Sugandhi and Prof. C. Muthamizhelvan for providing us the facilities and opportunity to carry out such research work under supervised work.

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